

Kanak Raj

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EDUCATION

- Birla Institute of Technology Mesra** Ranchi, India
 - Integrated Master's of Science - Mathematics & Computing; GPA: 8.88 2019 - 2024

PUBLICATIONS

- **Test-Time Scaling in the Wild: Why Exploitation, Not Exploration, Is the Bottleneck** arXiv 2026.
- **Transformers Remember First, Forget Last: Dual-Process Interference in LLMs** arXiv 2025.
- **K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries** at AAAI Spring Symposium 2024.
- **Multimodal Ingredient Substitution Knowledge Graph (MISKG) Dataset Release** 2024.
- **Cook-Gen: Robust Generative Modeling of Cooking Actions from Recipes** at IEEE SMC 2023.
- **Spatial Field Fusion Network (SFFNet) for Panoramic Dental X-ray Segmentation** at IEEE APSCON 2023.

ONGOING PROJECTS

- **Transformer State Tracking & Memory Interference** — Probing and attention-head analysis show LLMs (15+ models) suppress the correct answer to context updates before the final layer; confirmed via LoRA fine-tuning and a from-scratch GPT-2-small run. Under review at AAAI 2027.
- **Metacognitive Confidence in LLMs** — Using probing, activation steering, and patching, found sycophancy is encoded intrinsically at the same position that carries the model's confidence signal, prior to verbalization.

EXPERIENCE

- Thomson Reuters Lab** Bangalore, India
 - Applied Research Scientist Aug 2024 - Jun 2026
 - **LLM Reasoning & Reward Modeling:**
 - * Fine-tuned Qwen3.5-9B via Dr. GRPO for multi-turn legal citation retrieval (1,139 Swiss law queries): designed 3-tool agentic environment (BM25 search, article verification, submit) through 29-experiment ablation — 7-tool design collapsed to F1 0.066 vs. F1 0.244 with 3 tools
 - * Designed multi-stage reward: bootstrap stage rewards search behavior and citation format; optimize stage uses F1 with query quality, anti-hallucination, and over-prediction penalties
 - * Built IRAC (Issue-Rule-Application-Conclusion) Knowledge Graph from Westlaw corpus for DPO fine-tuning preference data; improved LegalBench benchmark performance
 - **Test-Time Scaling:**
 - * Implemented production module supporting Best-of-N, DVTS, Beam Search, and Particle Filtering steered by fine-tuned Bradley-Terry reward models (outcome and process)
 - * Conducted the first systematic study of test-time scaling on non-verifiable tasks (arXiv 2026): compute-matched (inference-FLOPs) benchmarking of 6 TTS paradigms across 5 open-ended QA domains (legal, medical, finance, instruction-following, writing); ~8,500 samples spanning deterministic to rubric-only verifiability
 - **Agentic Systems:**
 - * Architected multi-agent system for automated legal document editing with error-resolving agents and multi-turn reasoning; achieved 98% accuracy, 65% reduction in manual revision time, sub-8s latency per section across hundreds of documents monthly
- Microsoft Research** Bangalore, India
 - Research Intern (LLMs & AI4Code) Jan 2024 - Jul 2024
 - Built PwR Studio: NL→DSL translation system enforcing symbolic compilation checks on LLM outputs; reduced hallucinations by 85%, valid code generation 65%→95%; featured in Satya Nadella's keynote, adopted by 15+ organizations
 - Architected Jugalbandi platform converting DSL into FSM-based chatbot applications; 80% reduction in development time; enabled government agencies and NGOs to deploy multilingual voice/text bots across WhatsApp, Telegram, and web; adopted by Bhashini for national chatbot infrastructure
 - Open-sourced: PwR-Studio, PwR-NL2DSL, Jugalbandi Studio Engine
- AI Institute, University of South Carolina** Remote
 - Research Intern (Neuro-Symbolic AI & NLP) Dec 2022 - Apr 2024
 - Master's thesis with Dr. Amit Sheth: built Multimodal Ingredient Substitution Knowledge Graph (MISKG) — 27K ingredients, 80K substitution pairs, sourced from FlavorDB, ConceptNet, and Recipe1M; largest ingredient substitution corpus; adopted at UC Irvine + Stanford Health Hackathon 2024
 - Formulated cross-modal recipe retrieval and cooking action recognition; 95% recall; led to Cook-Gen (SMC'23)
- University of Maryland Baltimore** Remote
 - Research Intern (Natural Language Processing) Oct 2022 - Apr 2023
 - Developed personalized response generation models with Prof. Manas Gaur using reward scaling over BART & T5; improved NUBIA score by 10%; led to K-PERM (AAAI Spring Symposium 2024)